Intermediate Elective

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Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(lubridate)
library(dplyr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Figure planning

Final visual goal: compare the abundances of the most common inverts found at ncos in 2024

Create data frame

```r
invert_2024 <- aq_ins |>
  clean_names() |>
  rename(date = date_on_vial) |>
  filter(year(date) == 2024) |>
  select(ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode) |>
  pivot_longer(cols = ostracod:nematode,
               names_to = "taxon",
               values_to = "abundance") |>
  mutate(
```

```r
    abundance = replace_na(abundance, 0),
    taxon = str_to_title(str_replace_all(taxon, "_", " "))
  ) |>
  group_by(taxon) |>
  summarise(total_abundance = sum(abundance), .groups = "drop")
```

Plot

```r
library(tidyverse)
library(patchwork)

invert_2024_plot <- invert_2024 |>
  mutate(
    abundance_pct = total_abundance / sum(total_abundance),
    pct_scaled = 10 - (abundance_pct * 10),
    pct_label = paste0(round(abundance_pct * 100, 1), "%")
  )

invert_list <- list()

for(i in 1:nrow(invert_2024_plot)){
  invert_list[[i]] <- tibble(
    taxon = invert_2024_plot$taxon[i],
    x = c(0, 10, 10, 0),
    y = c(10, 10, rep(invert_2024_plot$pct_scaled[i], 2)),
    pct_label = invert_2024_plot$pct_label[i]
  )
}

line_data <- tibble(
  x = 0,
  xend = 10,
  y = 10,
  yend = 10
)

main_plot <-
  ggplot(data = bind_rows(invert_list)) +
  geom_segment(data = line_data,
```

```r
               aes(x = x, xend = xend, y = y, yend = yend),
               linewidth = 2,
               color = "#cdcdc0",
               inherit.aes = FALSE) +
  geom_polygon(aes(x = x,
                   y = y,
                   group = taxon),
               fill = "#0c1931") +
  geom_text(aes(x = 5, y = 0, label = taxon),
            size = 3,
            family = "sans",
            fontface = "bold",
            color = "#626d71") +
  geom_text(aes(x = 5, y = 3, label = pct_label),
            size = 3,
            family = "sans",
            fontface = "bold",
            color = "black") +
  facet_wrap(~taxon, nrow = 2) +
  coord_polar() +
  ylim(0, 10) +
  labs(title = "Aquatic Invertebrate Abundance in 2024") +
  theme_void() +
  theme(
    strip.background = element_blank(),
    strip.text = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold",
                              size = 24, color = "black", family = "sans"),
    panel.background = element_rect(fill = "#f4f2f2", color = NA),
    plot.background = element_rect(fill = "#f4f2f2", color = NA),
    plot.margin = margin(5, 0, 0, 0)
  )

legend_data <- tibble(
  type = c(10, 25, 50, 75),
  type_scales = c(1, 2.5, 5, 7.5),
  label = paste0(type, "%"),
  y = 10 - type_scales
)

legend_list <- list()
```

```
for(i in 1:nrow(legend_data)){
  legend_list[[i]] <- tibble(
    label = legend_data$label[i],
    type = legend_data$type[i],
    x = c(0, 10, 10, 0),
    y = c(10, 10, rep(legend_data$y[i], 2))
  )
}

legend_plot <-
  ggplot(data = bind_rows(legend_list)) +
  geom_segment(data = line_data,
               aes(x = x, xend = xend, y = y, yend = yend),
               linewidth = 2,
               color = "#cdcdc0",
               inherit.aes = FALSE) +
  geom_polygon(aes(x = x,
                   y = y,
                   group = type),
               fill = "#0c1931") +
  facet_wrap(~label, nrow = 1) +
  coord_polar() +
  ylim(0, 10) +
  theme_void() +
  theme(
    strip.text = element_text(face = "bold", color = "#0c1931",
                              family = "sans"),
    panel.background = element_rect(fill = "#f4f2f2", color = NA),
    plot.background = element_rect(fill = "#f4f2f2", color = NA),
    plot.margin = margin(t = 0, r = 0, b = 0, l = 30)
  )

title_plot <-
  ggplot() +
  geom_text(aes(x = 0, y = 2, label = "Percentage of Total Invertebrate Abundance"),
            family = "sans",
            fontface = "bold",
            hjust = 0,
            size = 3) +
  geom_text(aes(x = 0, y = 1,
                label = "The thickness of the inner perimeter of each circle\nrepresents the
            family = "sans",
```

```
             hjust = 0,
             size = 2) +
  ylim(0, 5) +
  scale_x_continuous(limits = c(0, 3), expand = c(0, 0)) +
  theme_void() +
  theme(
    panel.background = element_rect(fill = "#f4f2f2", color = NA),
    plot.background = element_rect(fill = "#f4f2f2", color = NA),
    plot.margin = margin(0, 0, 0, 0)
  )

layout <- c(
  area(t = 1, l = 1, b = 18, r = 15),
  area(t = 19, l = 1, b = 20, r = 5),
  area(t = 16, l = 6, b = 20, r = 15)
)

final_plot <- main_plot + legend_plot + title_plot +
  plot_layout(design = layout)

print(final_plot)
```